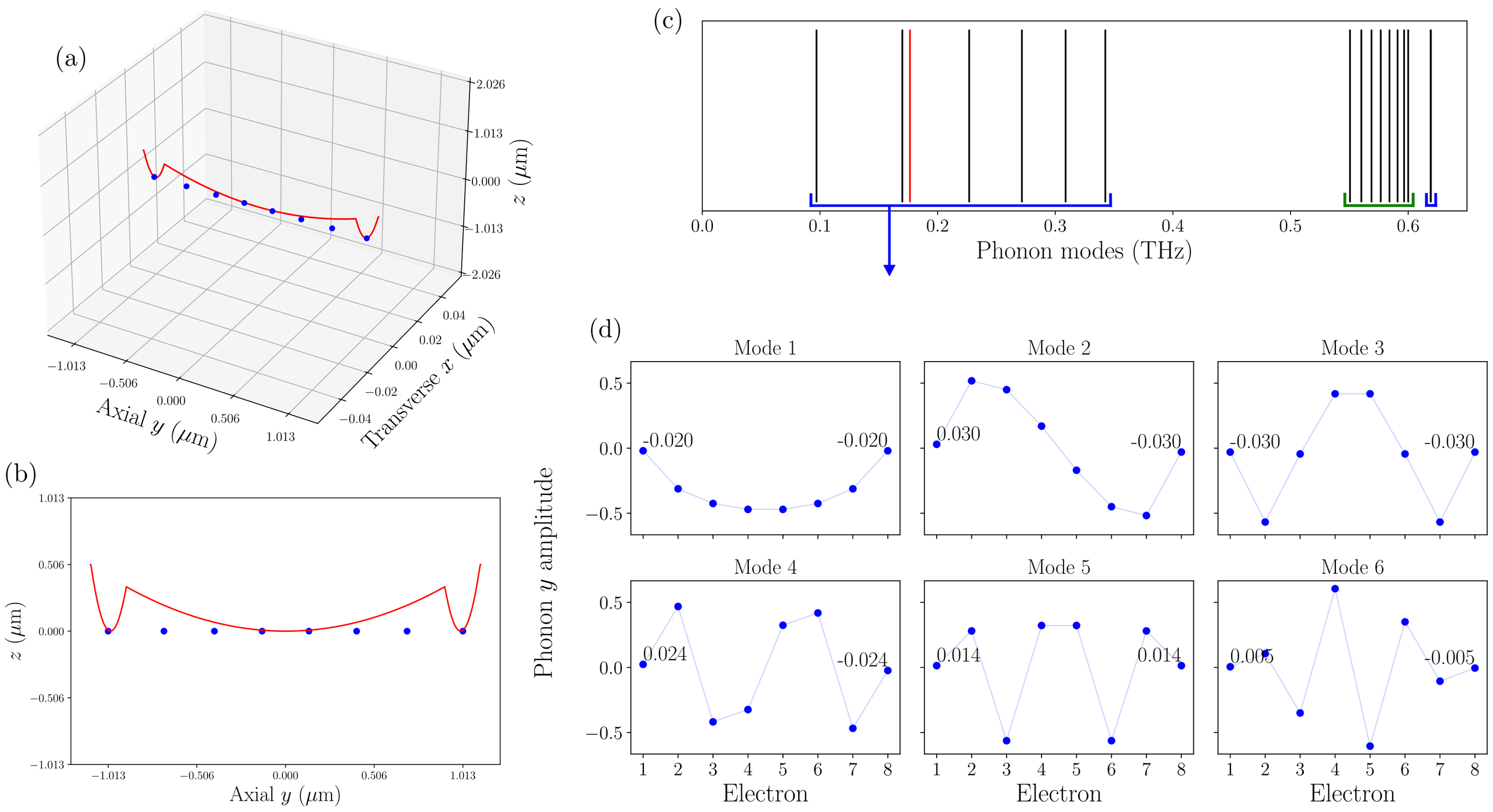

(a)
2.026
1.013
0.000
−1.013
−2.026
z (μm)
−1.013
−0.506
0.000
0.506
1.013
Axial y (μm)
0.04
0.02
0.00
−0.02
−0.04
Transverse x (μm)
(b)
1.013
0.506
0.000
−0.506
−1.013
z (μm)
−1.013
−0.506
0.000
0.506
1.013
Axial y (μm)
(c)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Phonon modes (THz)
(d)
Mode 1
Mode 2
Mode 3
Mode 4
Mode 5
Mode 6
Phonon y amplitude
0.5
0.0
−0.5
-0.020
-0.020
0.030
-0.030
-0.030
-0.030
0.024
-0.024
0.014
0.014
0.005
-0.005
1
2
3
4
5
6
7
8
Electron